**AKASHDIP MAHAPATRA | TECHNICAL PORTFOLIO ADDENDUM**

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**ENTERPRISE SYSTEM ARCHITECTURE**

**Project:** ODIE Serverless Data Pipeline (Aerospace IoT & Telemetry)

**Role:** Data Engineer / DevOps & Infrastructure Maintainer

**Scale:** Production-grade, high-throughput, multi-tiered AWS serverless architecture processing continuous real-time aviation telemetry and IoT data.

**System Architecture Overview:**

The system is a decoupled, event-driven data lake consisting of distinct ingestion, processing, storage, and distribution layers.

- **Ingestion: High throughput** streaming via Amazon MSK (Kafka) and AWS IoT Core.
- **Routing:** Heavy utilization of Amazon **EventBridge** to trigger asynchronous workflows based on object creation in S3 Raw Zones.
- **Compute:** Python-based transformation logic running in containerized AWS Lambda functions (managed via ECR).
- **Storage & Distribution:** Tiered S3 Data Lake (Raw, Processed, Curated), DynamoDB for real-time state management, and Amazon SNS / Kinesis Firehose for downstream data distribution.

**Key Engineering Responsibilities:**

- **CI/CD Pipeline Ownership:** Maintain highly structured GitHub Actions workflows (lambda-container-build-dev.yaml, terraform-release.yaml) handling the end-to-end build, testing, and deployment of Dockerized Lambda functions and Terraform modules across Dev, QA, UAT, and Prod environments.
- **Infrastructure as Code (IaC):** Manage complex infrastructure using modular Terraform, defining VPCs, IAM roles with strict least-privilege policies, S3 event triggers, and Lambda resource provisioning.
- **Container Security & Maintenance:** Monitor Dockerfiles and requirements.txt dependencies across multiple microservices (e.g., *AIDX Sequencer, Aircraft Towing Processor, GSE Tracking*), actively identifying and patching CVEs to maintain strict enterprise security standards.
- **Observability & Troubleshooting:** Utilize AWS CloudWatch and Datadog for system-wide tracing, log aggregation, and error handling (managing Dead Letter Queues) to ensure zero data loss in the ingestion layer.

## Hardware & Academic Engineering

**APPLIED NETWORKING & IOT ENGINEERING (Selected Projects)**

**1. Network Attached Storage (NAS) & Custom Web Hosting via ESP32**

- **Tech Stack:** ESP32, Arduino, HTTP, Local Area Network (LAN).
- **Details:** Engineered a microcontroller-based NAS to serve files over Wi-Fi. Extended this architecture to host a custom HTML web interface directly on the ESP32 for localized hardware control, simulating a micro web server environment.

**2. Wi-Fi Penetration & Network Security Tool**

- **Tech Stack:** ESP8266/ESP32, Python, Packet Sniffing.
- **Details:** Designed a wireless network testing system capable of scanning 802.11 traffic, capturing handshake packets, and broadcasting de-authentication packets to test network access point vulnerabilities.

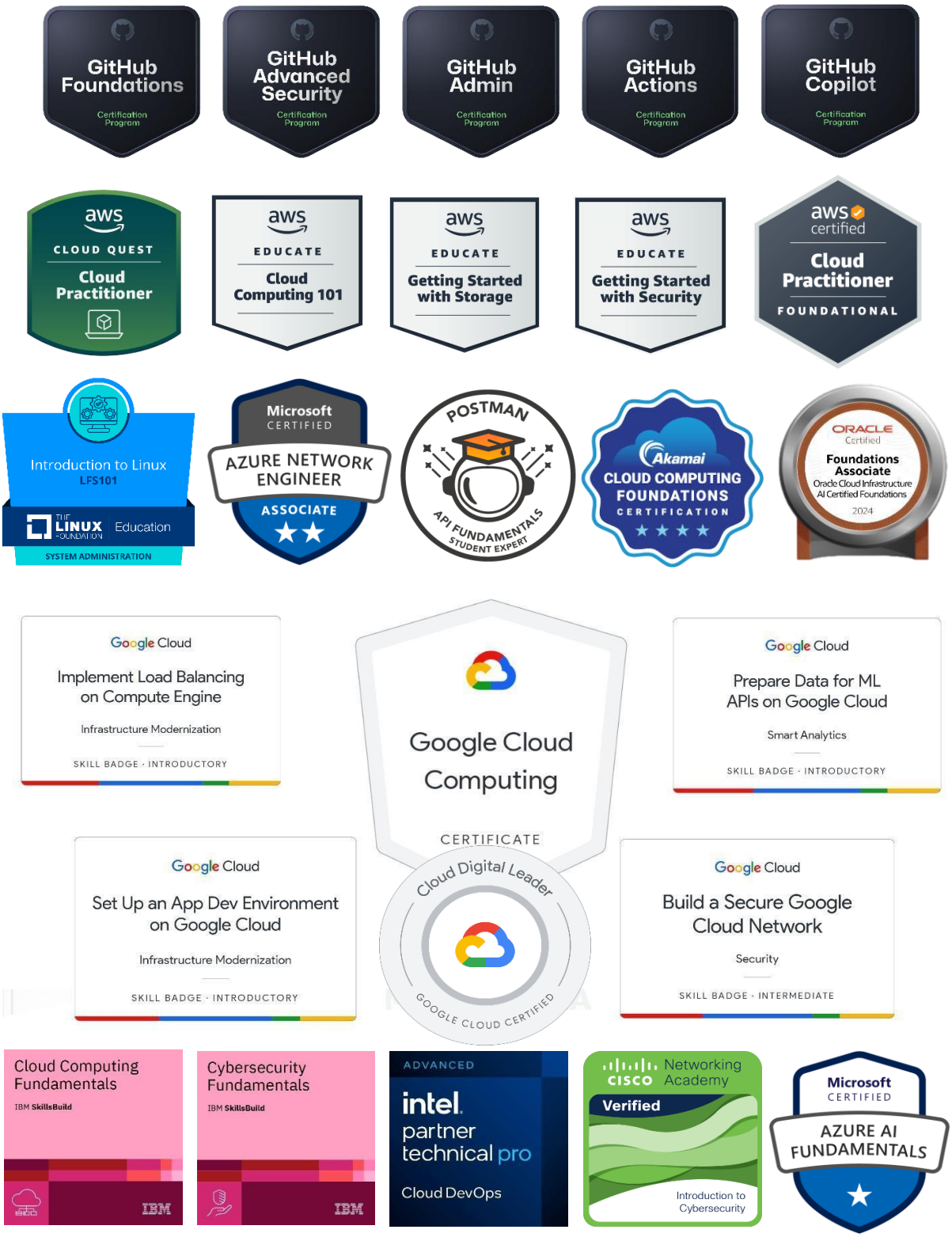
**3. Real-Time GPS Tracking & Telemetry System**

- **Tech Stack:** Node.js, Express, Socket.IO, ESP32, GPS Modules.
- **Details:** Built an end-to-end IoT telemetry pipeline. Captured live GPS coordinates on edge hardware (ESP32) and streamed the data via WebSockets to a Node.js backend for real-time mobile visualization.

**4. Lightweight Python Web Server (From Scratch)**

- **Tech Stack:** Python, TCP Sockets, OSI Model Layer 4/7.
- **Details:** Bypassed standard frameworks (like Flask/Django) to build a raw HTTP server using basic Python TCP socket programming. Handled low-level GET/POST request parsing and static content delivery to solidify core networking fundamentals.

## PROFESSIONAL CREDENTIALS & CERTIFICATIONS




**Verify in Credly -** https://www.credly.com/users/**akashdip2001**/

# GitHub Advanced Security

ISSUED TO

AKASHDIP MAHAPATRA

GitHub Advanced Security
Certification Program

Issued on: 10 FEB 2025 | Expires on: 10 FEB 2028 | Issued by: GitHub

Verify: https://www.credly.com/go/0uS9HRCw

Microsoft

# AKASHDIP MAHAPATRA

has successfully completed

## Describe Azure compute and networking services

28 April 2024

Satya Narayana Nadella

Microsoft

# AKASHDIP MAHAPATRA

has successfully passed all requirements for

## Microsoft Certified: Azure Network Engineer Associate

Credential ID: AA9E4130816324D7

Certification number: D498AD-9345B3

Earned on: 20 October 2025

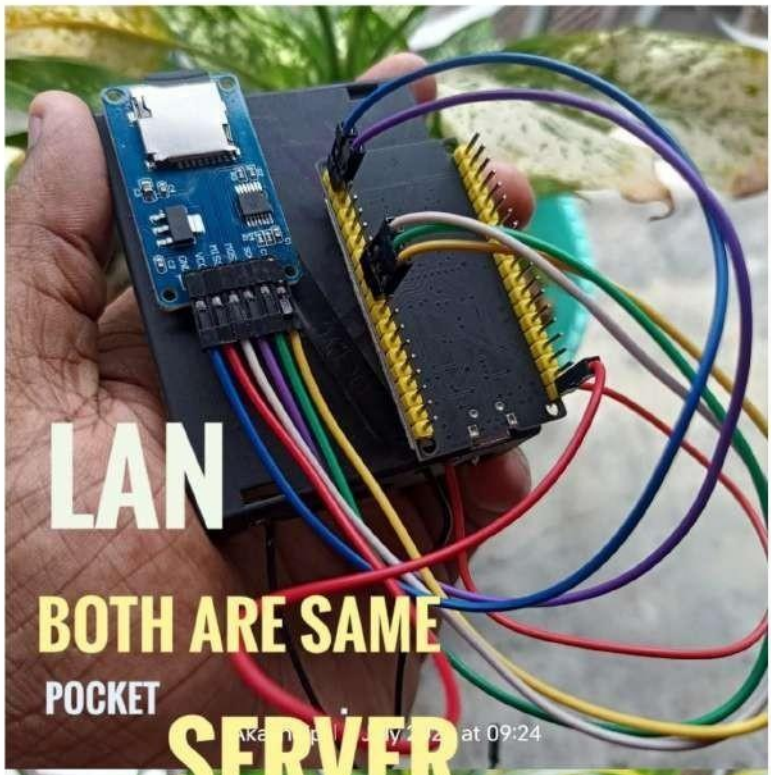
Expires on: 21 October 2026

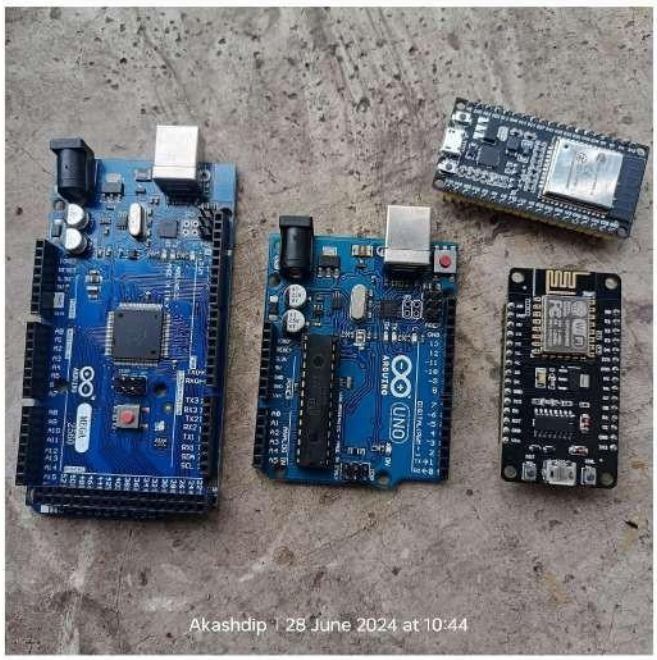
Online Verifiable

Microsoft CERTIFIED ASSOCIATE

Satya Narayana Nadella

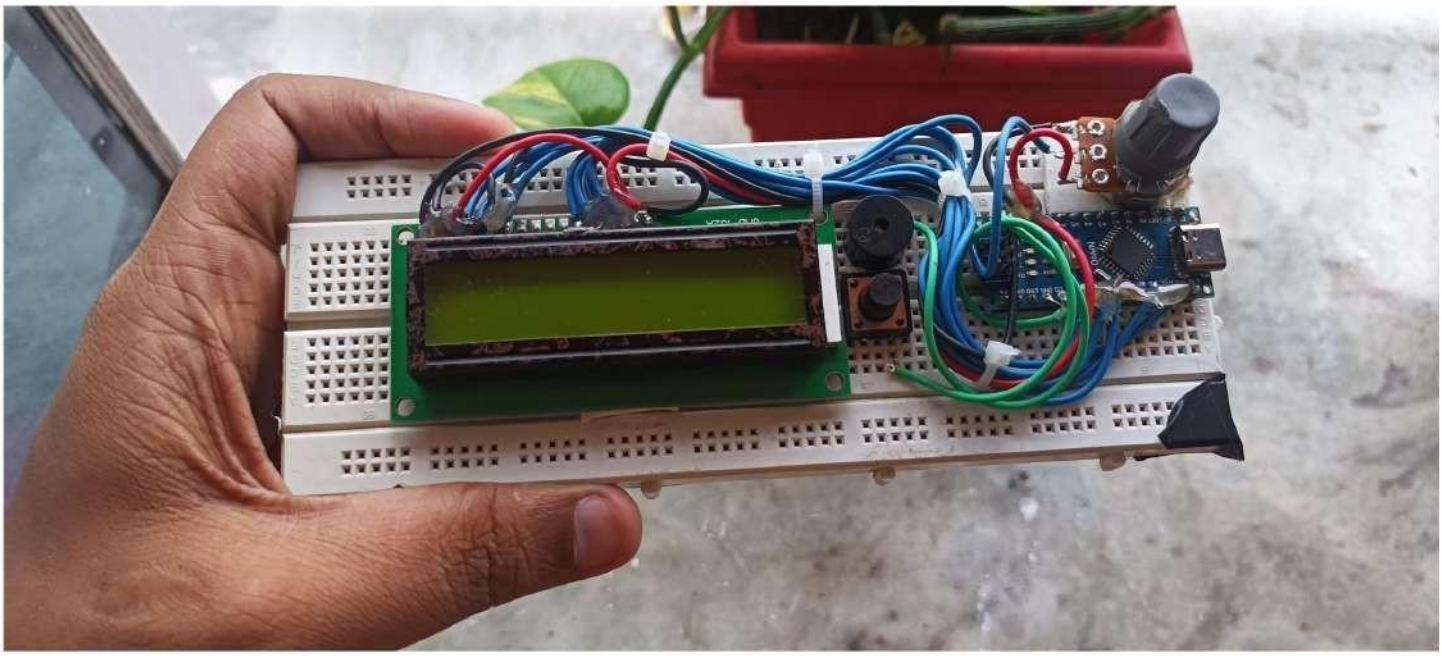
# DIY home CLOUD SERVER using ESP32 & Sd Card MODULE ... project research QR



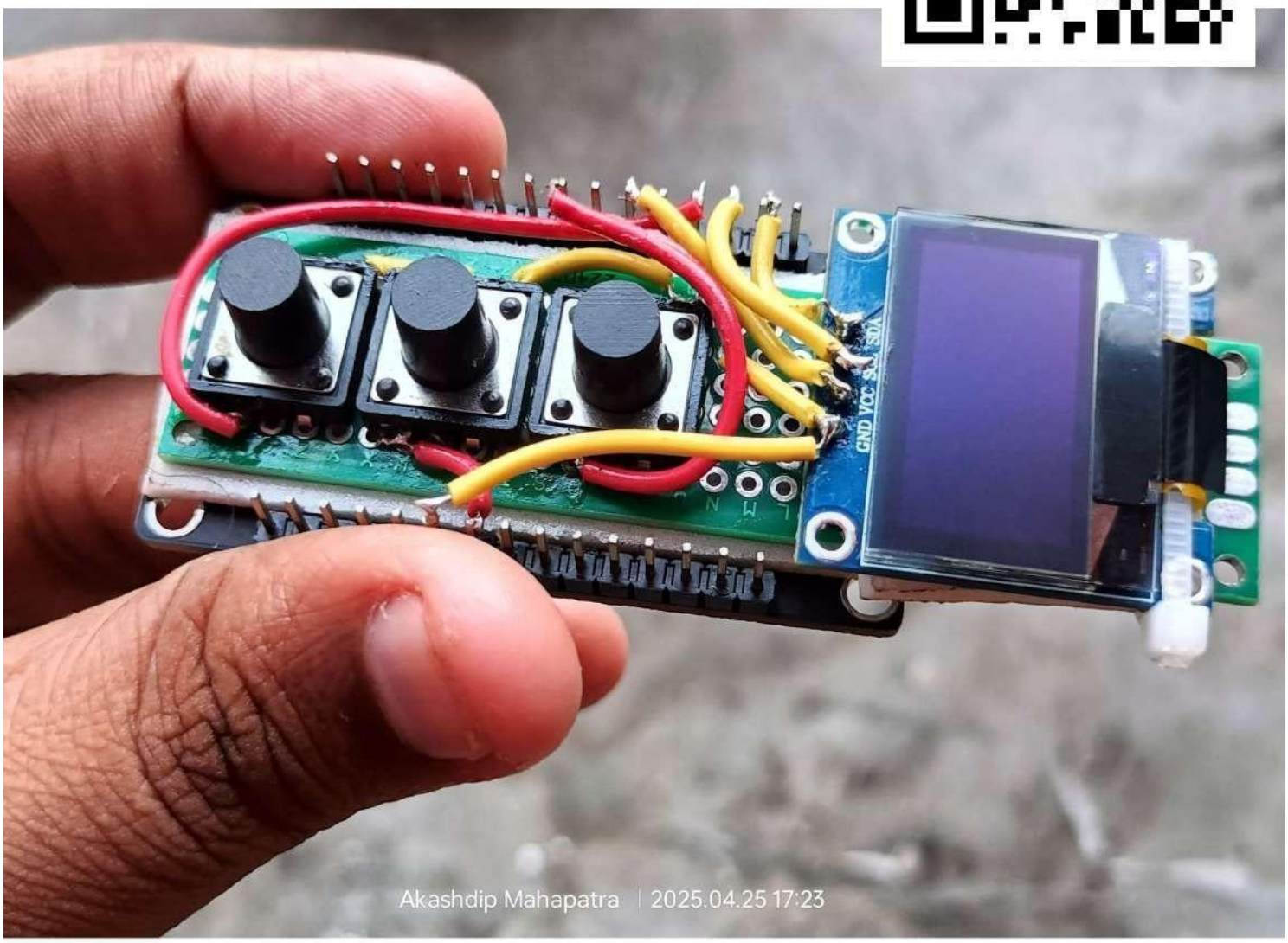
Google Cloud

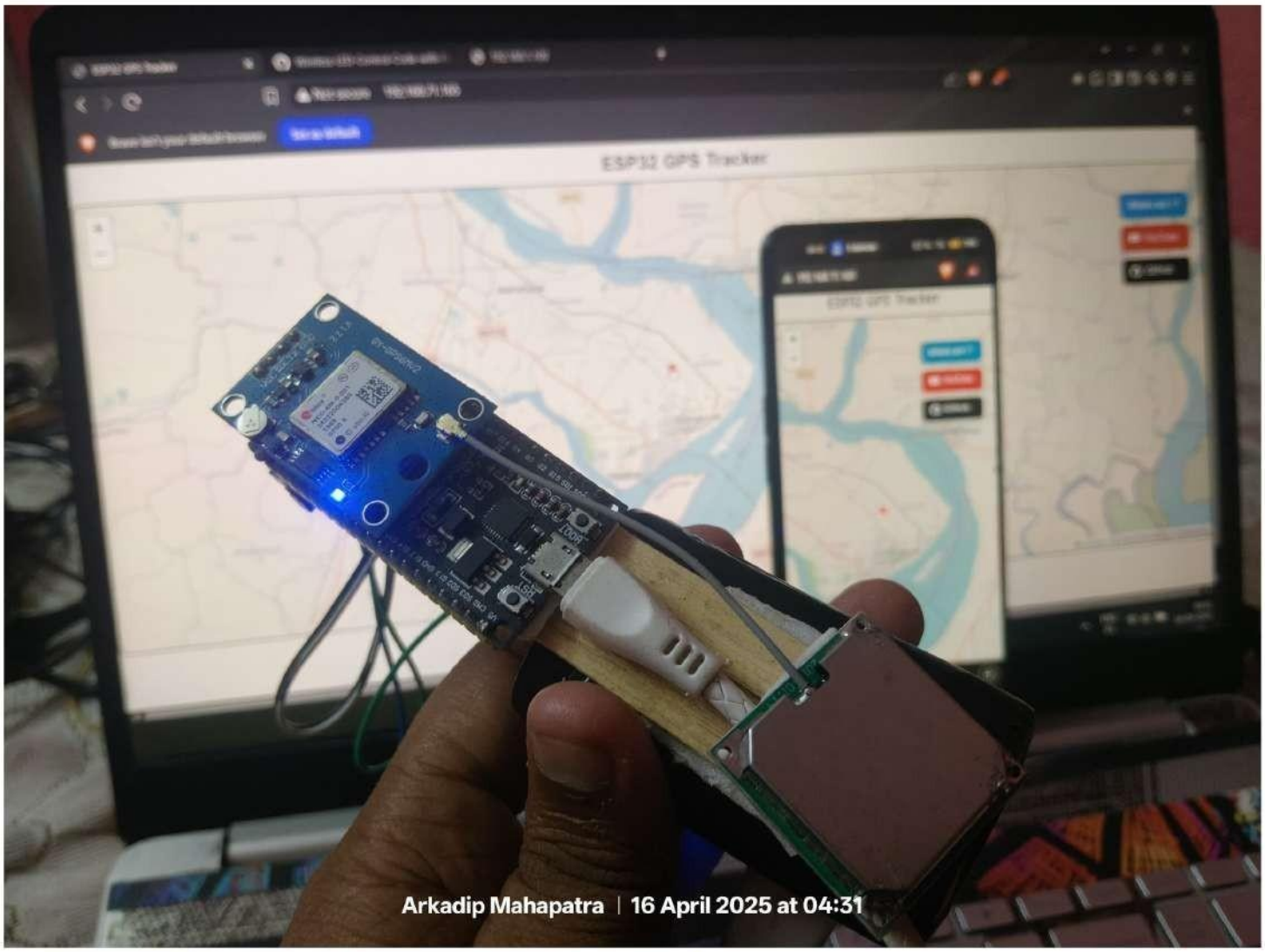
# Game with IOT

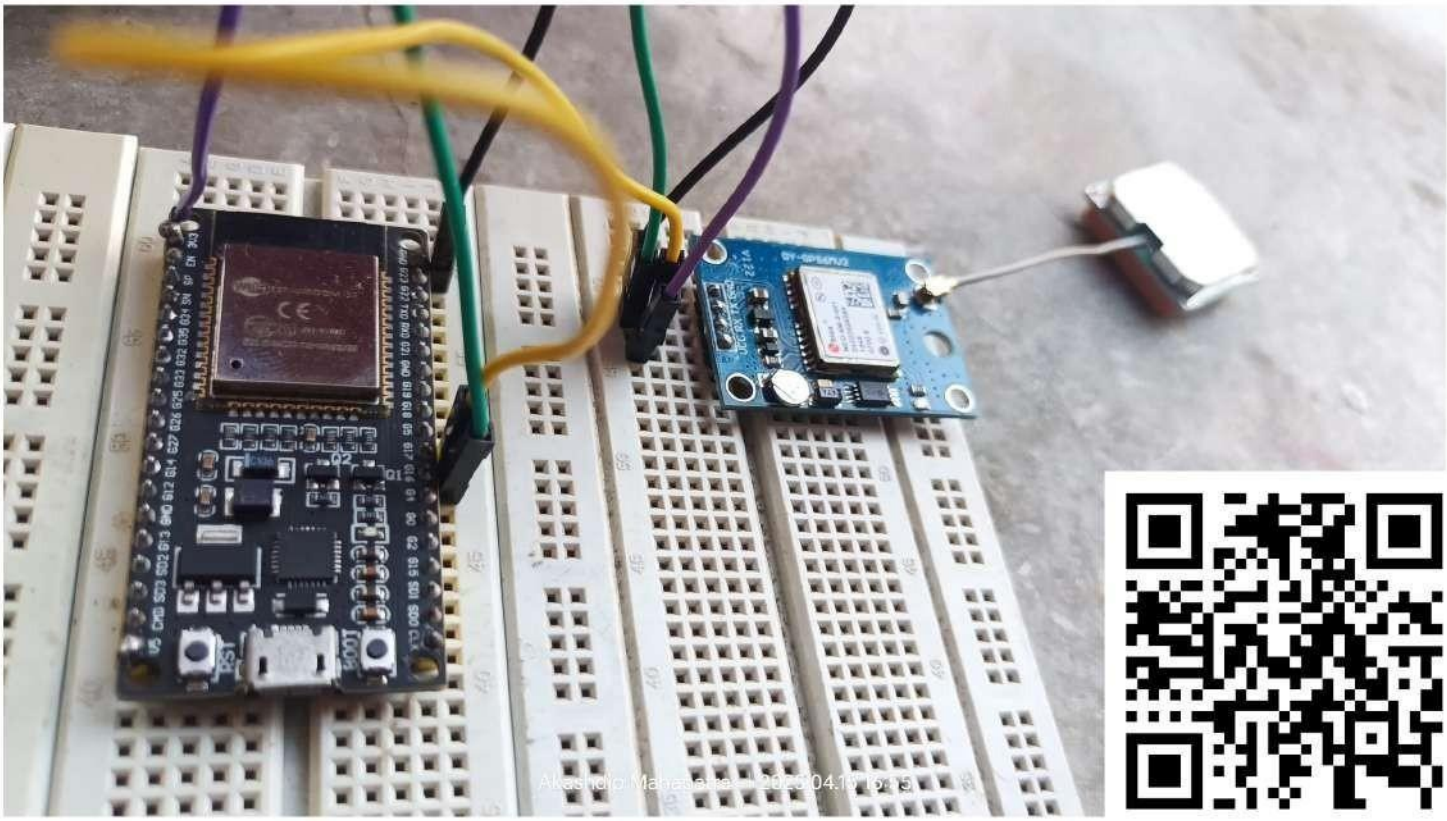


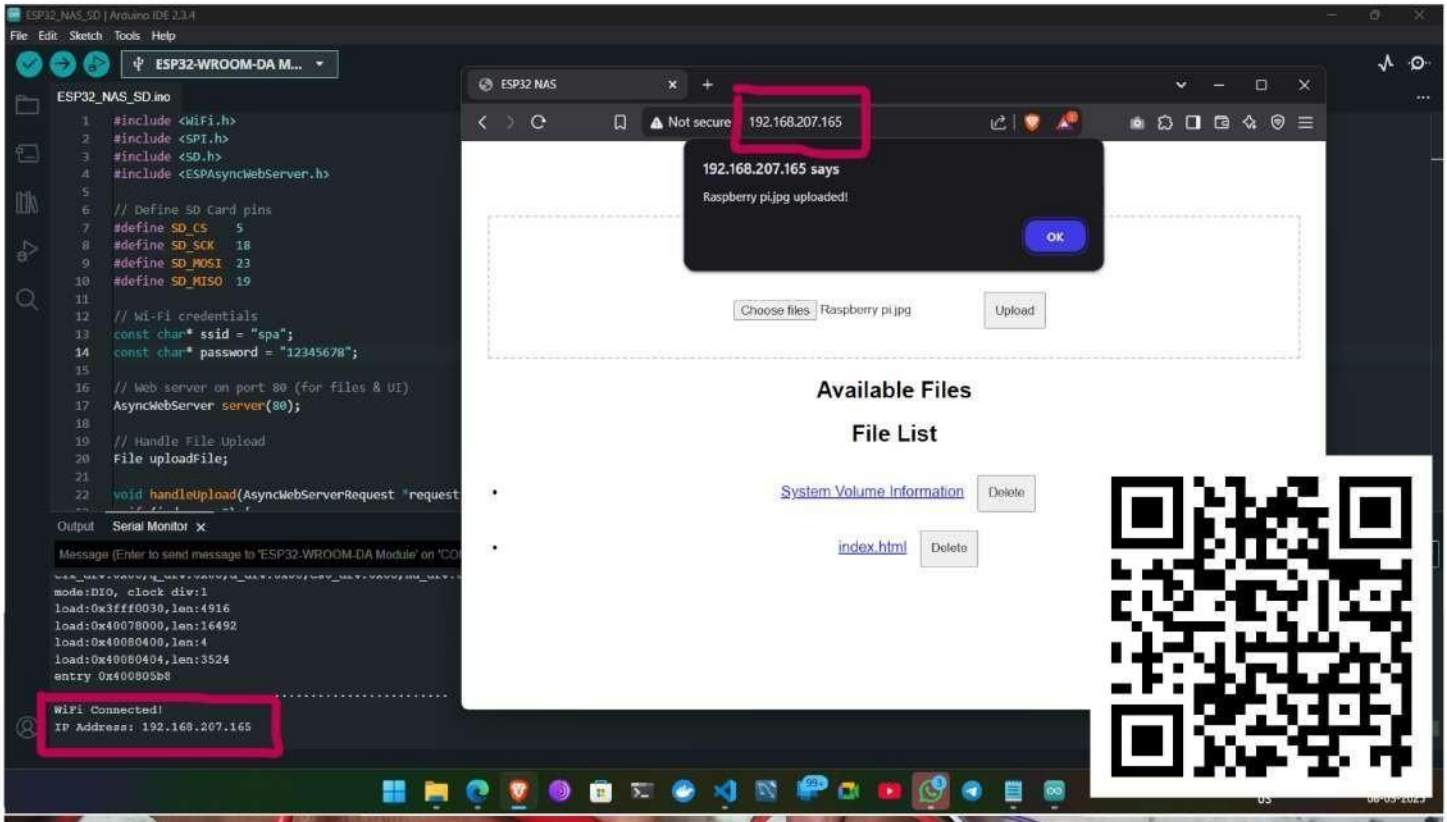
15:02
1 device
30%
GND VCC SCL SDA
SCAN
SELECT
ATTACK
PACKET MONITOR
CLOCK
Akashdip Mahapatra

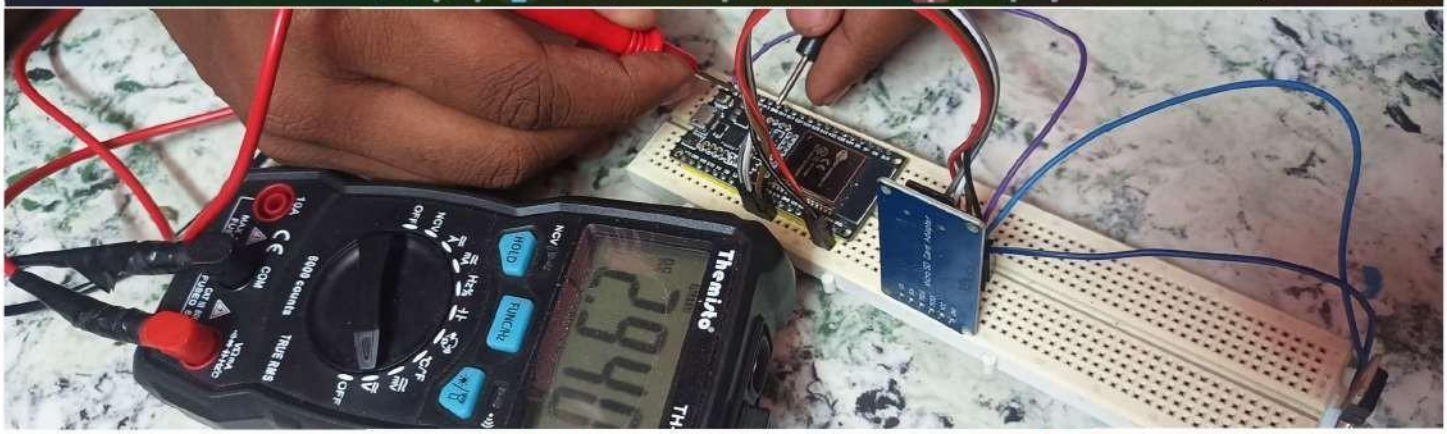
ESP8266 Deauther
OLED Mod
by Arkadip
v2.1.0
Akashdip Mahapatra

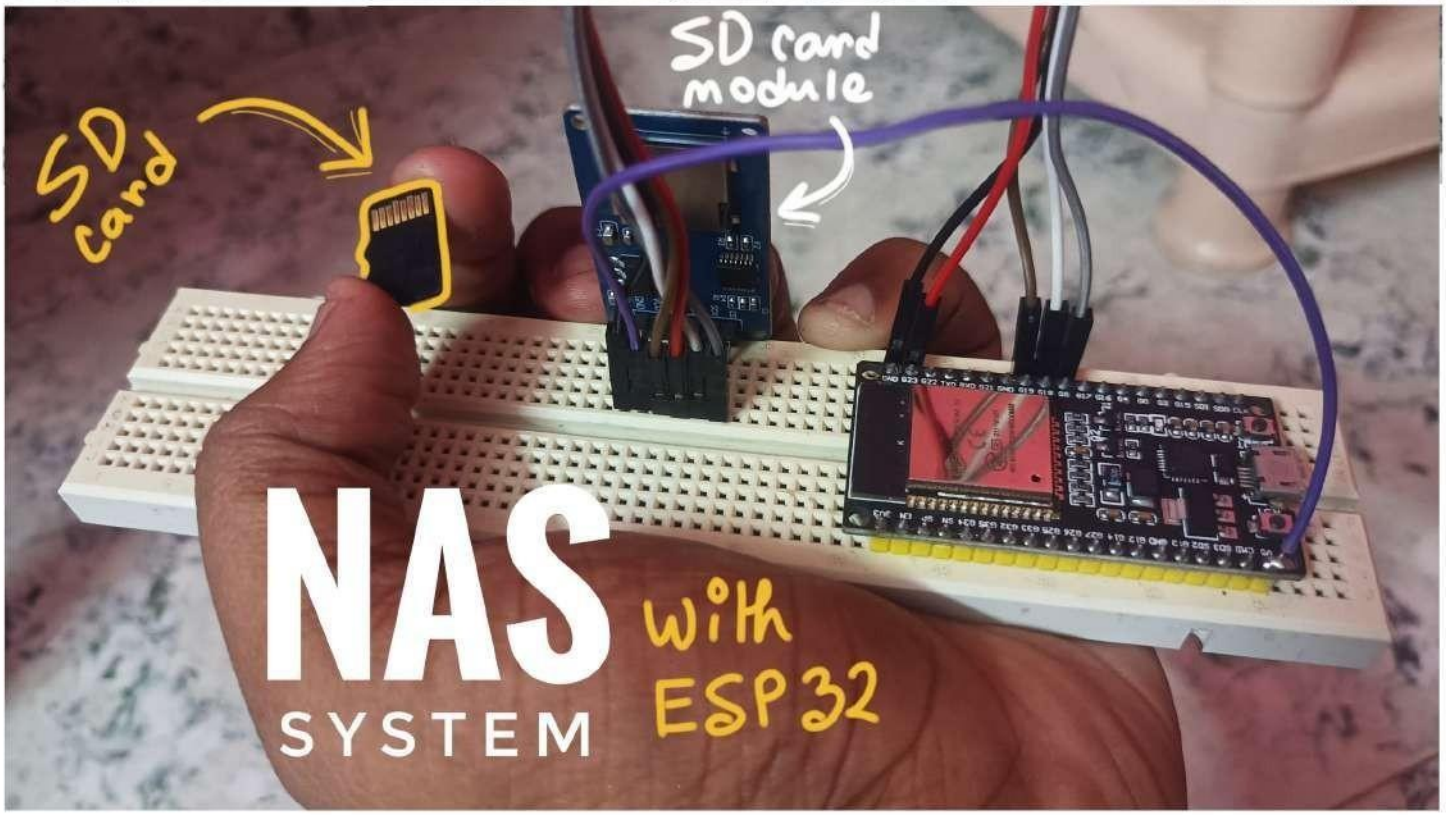

GND VCC SCL SDA
Akashdip Mahapatra | 2025.04.25 17:23


ESP32 GPS Tracker
Arkadip Mahapatra | 16 April 2025 at 04:31



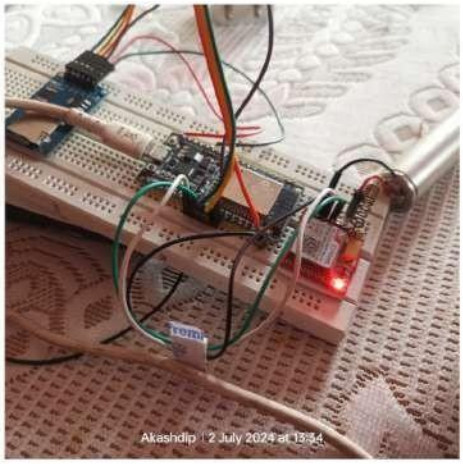

ESP32_NAS_SD | Arduino IDE 2.3.4
File Edit Sketch Tools Help
ESP32-WROOM-DA M...
ESP32_NAS_SD.ino
#include <WiFi.h>
#include <SPI.h>
#include <SD.h>
#include <ESPAsyncWebServer.h>
// Define SD Card pins
#define SD_CS 5
#define SD_SCK 18
#define SD_MOSI 23
#define SD_MISO 19
// Wi-Fi credentials
const char* ssid = "spa";
const char* password = "12345678";
// Web server on port 80 (for files & UI)
AsyncWebServer server(80);
// Handle File Upload
File uploadFile;
void handleUpload(AsyncWebServerRequest *request
Output Serial Monitor
mode:DIO, clock div:1
load:0x3fff0030,len:4916
load:0x40078000,len:16492
load:0x40080400,len:4
load:0x40080404,len:3524
entry 0x400805b8
WiFi Connected!
IP Address: 192.168.207.165
ESP32 NAS
Not secure 192.168.207.165
192.168.207.165 says
Raspberry pi.jpg uploaded!
OK
Choose files Raspberry pi.jpg
Upload
Available Files
File List
System Volume Information
Delete
index.html
Delete



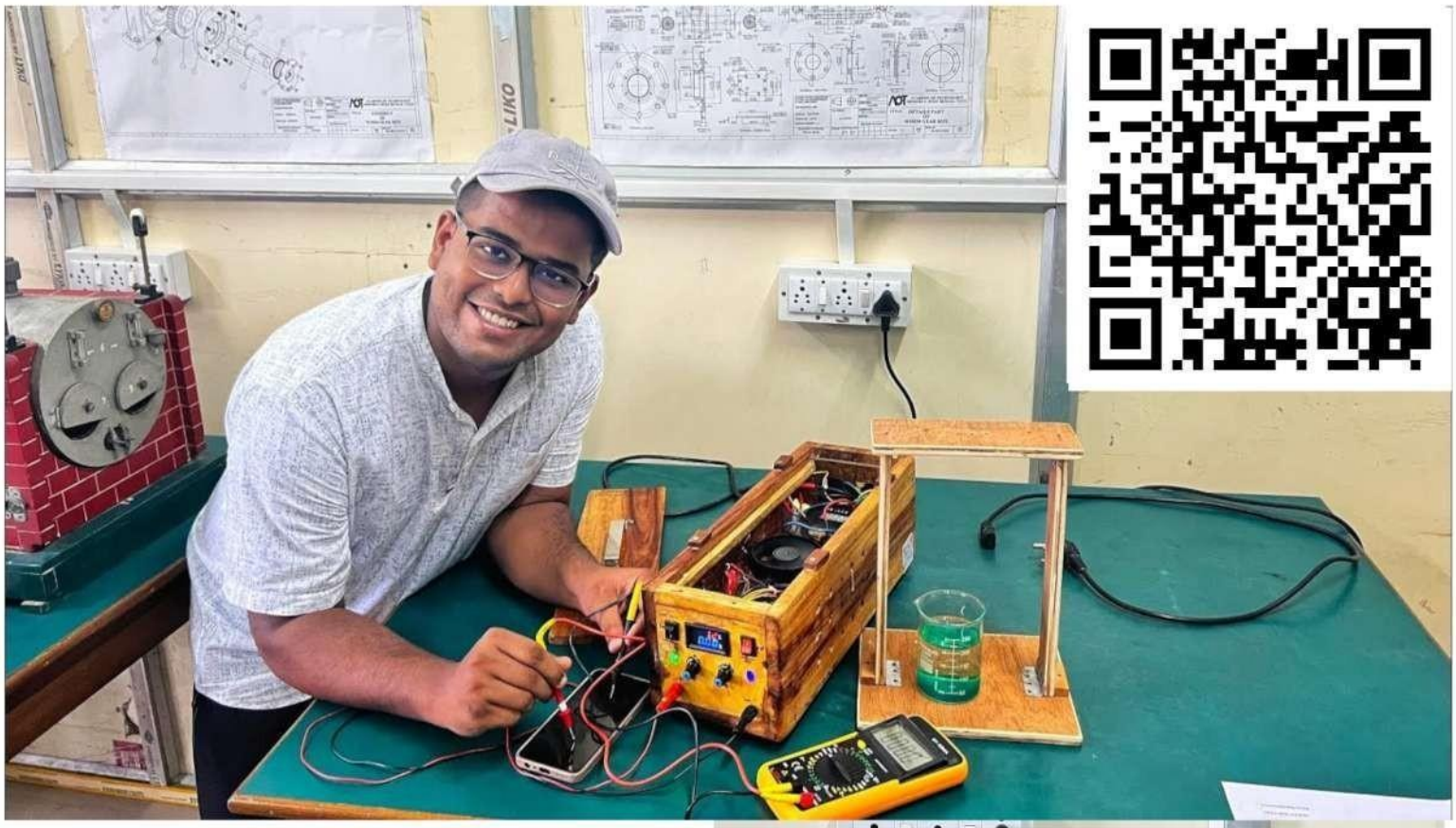

SD card
SD card module
NAS
SYSTEM
with
ESP32

ashdip | 2 July 2024 at 09:24

Akashdip | 1 July 2024 at 14:50

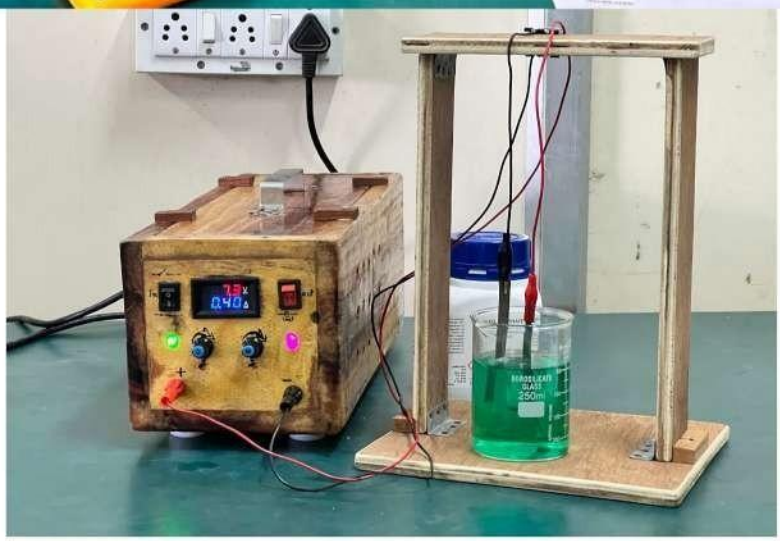

Akashdip | 2 July 2024 at 13:34

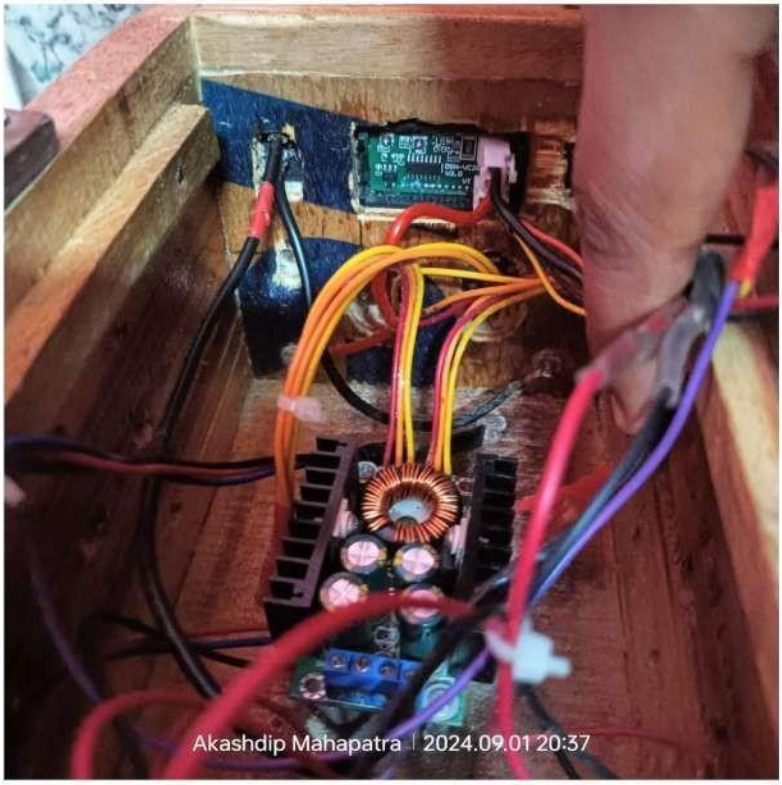
my
POCKET
SERVER
Akashdip | 1 July 2024 at 17:48
ESP32 + SC CARD



# final year project

**300 WATT variable POWER SUPPLY**
**MAX OUT-PUT 10A & 30 Volt**





Microsoft

# AKASHDIP MAHAPATRA

has successfully passed all requirements for

## Microsoft Certified: Azure AI Fundamentals

Credential ID: 3A9D77112C32F726

Certification number: DD95C6-W03C74

Earned on: June 18, 2025

Online Verifiable

Microsoft CERTIFIED FUNDAMENTALS

Satya Narayana Nadella

**AWS Certified Cloud Practitioner**

**Notice of Exam Results**

| | |
|---|---|
| Candidate: AKASHDIP MAHAPATRA | Exam Date: 6/22/2025 |
| Candidate ID: AWS04771330 | Registration Number: 509483638 |
| Candidate Score: 1000 | Pass/Fail: PASS |

**Congratulations! You have successfully completed the AWS Certified Cloud Practitioner and you are now AWS Certified.**

The AWS Certified Cloud Practitioner (CLF-C02) has a scaled score between 100 and 1,000. The scaled score needed to pass the exam is 700.

As a reminder, you are bound by the Candidate Code of Conduct you accepted when taking your exam, and you are expected to keep the contents of the examination secure. If you have any questions or require assistance, please contact us.

Congratulations again on your achievement and the ability to claim the associated benefits, such as digital badges and discount vouchers for future exams. To take advantage of the benefits associated with your AWS Certified Cloud Practitioner certification, explore the "Benefits" section of your AWS Certification Account.

Thank you,

AWS Training and Certification